MARKET BASKET ANALYSIS PROJECT

OBJECTIVE:

The primary objective of the Market Basket Analysis System is to analyze customer transaction data to discover meaningful associations and co-occurrence patterns among products, enabling data-driven decisions for product placement, cross-selling, and promotional strategies.

1. To automate the extraction of frequent itemsets from large transactional datasets.

2. To generate interpretable association rules using support, confidence, and lift.

3. To provide a software system that transforms raw sales data into actionable business insights.

FUNCTIONS:

load_dataset() - Reads transaction data from CSV / database.

clean_data() - Removes null, removes duplicate transactions, standardizes item names

create_transactions() - Converts rows into a list of itemsets.

encode_transactions() - Transforms transactions into a binary matrix (one-hot encoding)

generate_frequent_itemsets() - Finds frequent itemsets.

generate_association_rules() - Generates rules from frequent itemsets.

filter_rules() - Removes misleading or weak rules.

SCOPE :

1. Analysis of transactional retail or e-commerce data.

2. Discovery of product association patterns.

3. Support for marketing strategies like bundling and promotions.

4. Small to medium-scale datasets suitable for academic environments.

CONSTRAINTS :

Technical Constraints

• Performance degradation with very large datasets (especially Apriori).

• High memory consumption for dense itemsets.

• Static datasets (no real-time updates).

Data Constraints

• Quality of results depends heavily on data cleanliness.

• Sparse data can lead to weak or misleading rules.

• Requires sufficient transaction volume for meaningful patterns.

Project Constraints

• Limited development time.

• Restricted computational resources.

• Academic-level dataset availability.

SDLC MODEL USED: Waterfall Model

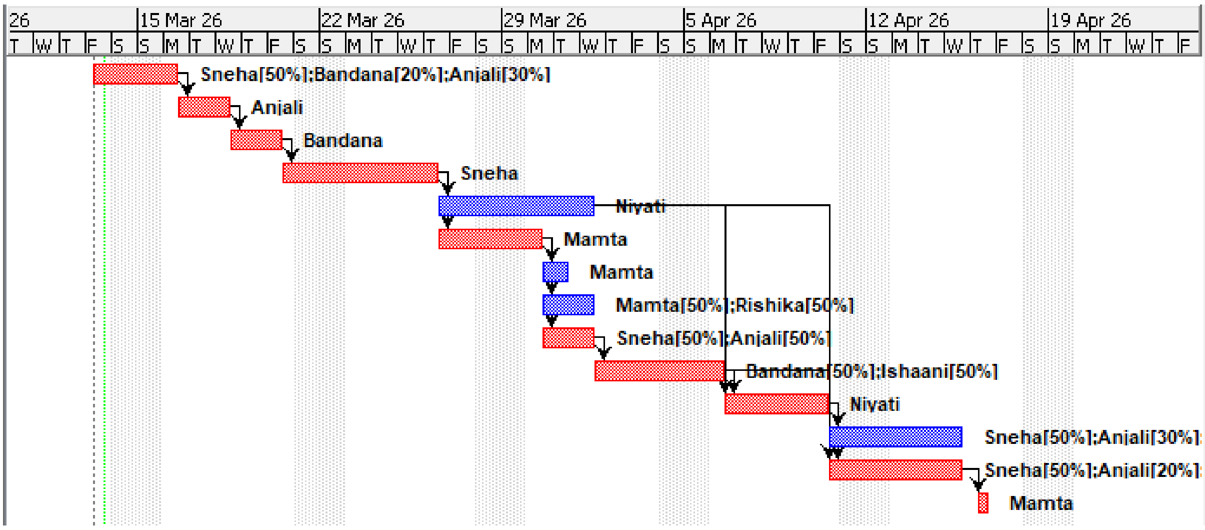
The Market Basket Analysis System follows the Waterfall Model because the requirements were clearly defined before development began. The project was developed through sequential phases including Requirement Analysis, System Design, Implementation, Testing, and Documentation. Since the functionality of frequent itemset generation, support calculation, confidence calculation, and association rule generation was predetermined and unlikely to change during development, the Waterfall Model was considered suitable.

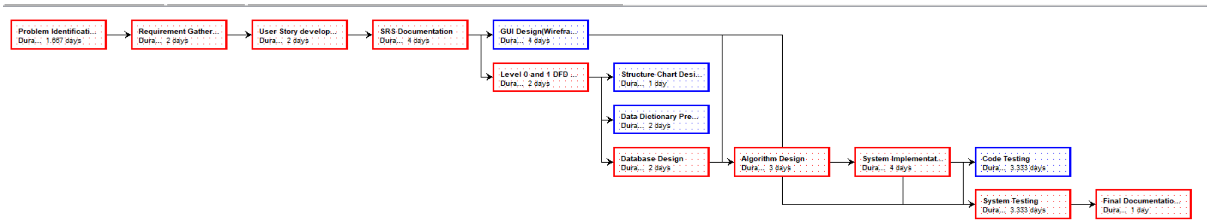
USER ID	USER STORY	PRIORITY	RISKS	STORY POINTS
1	As a Buisness owner, I want to know the best selling cominations so that I can bundle them together.	Must Do	High	6
2	As a Buisness owner, I want information about the newly emerging buying patterns so that we can act on them.	Should Do	Moderate	3
3	As a buisness owner, I want to compare association results across different timelines so that I can plan sales.	Should Do	Low	4
4	As a buisness owner, I want reccomendations ranked by profits so that I don’t waste time on low value associations.	Could Do	Moderate	5
5	As a buisness owner, I want montly buying patterns so that I can plan promotions ahead of time.	Could Do	Moderate	2
6	As a buisness owner, I want a summary dashboard of the sales so that I can understand insights at a glance.	Could Do	Low	4
7	As a store manager, I want to know which products are purchased ferequently so that I can place them nearby on shelves.	Must Do	High	5
8	As a store manager, I want downloadable report so that I can share that with everyone.	Should Do	Low	3
9	As a store manager, I want alerts for frequetly bought-together itrems that are placed far apart so that layout issues can be corrected.	Must Do	High	6
10	As a store manager, I want indicators about current trends so that we can anticipate demand changes and prepare beforehand.	Should Do	High	5
11	As a store manager, I want to exclude out of stock products from analysis so that results stay realistic.	Could Do	Moderate	2
12	As a store manager, I want to analyze buying patterns by store section so that I can optimize local layouts.	Should Do	Moderate	3
13	As a marketing executive, I want to know ongoing trends so that we can promote them.	Must Do	Low	3
14	As a marketing executive, I want to identify add-on products so that we can provide chechout-time reccomendations.	Should Do	High	5
15	As a marketing executive, I want to know the seasonal combinations so that we can plan to promote them beforehand.	Could Do	Low	4

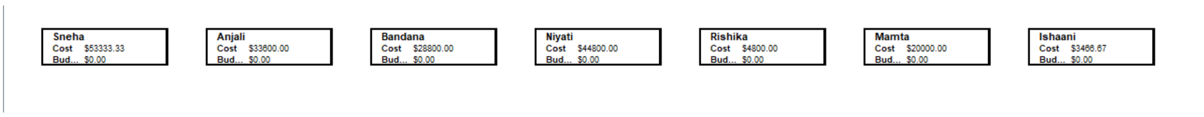
USER ID	SPRINT	USER STORY	STORY POINTS	PRIORITY	RISK	USER ID on which dependent
7	1	As a store manager, I want to know which products are purchased ferequently so that I can place them nearby on shelves.	5	Must do	High	
8	1	As a store manager, I want downloadable report so that I can share that with everyone.	3	Should do	Low	
11	1	As a store manager, I want to exclude out of stock products from analysis so that results stay realistic.	2	Could do	Moderate	7
12	1	As a store manager, I want to analyze buying patterns by store section so that I can optimize local layouts.	3	Should do	Moderate	
1	1	As a Buisness owner, I want to know the best selling cominations so that I can bundle them together.	6	Must do	High	8
		SPRINT 1 TOTAL STORY POINTS - 19				
2	2	As a Buisness owner, I want information about the newly emerging buying patterns so that we can act on them.	4	Should do	Low	8
5	2	As a buisness owner, I want montly buying patterns so that I can plan promotions ahead of time.	2	Could do	Moderate	7
13	2	As a marketing executive, I want to know ongoing trends so that we can promote them.	3	Must do	Low	7
14	2	As a marketing executive, I want to identify add-on products so that we can provide chechout-time reccomendations.	5	Should do	High	7
		SPRINT 2 TOTAL STORY POINTS - 14				
6	3	As a buisness owner, I want a summary dashboard of the sales so that I can understand insights at a glance.	4	Could do	Low	8
3	3	As a buisness owner, I want to compare association results across different timelines so that I can plan sales.	4	Should do	Low	6
15	3	As a marketing executive, I want to know the seasonal combinations so that we can plan to promote them beforehand.	4	Could do	Low	6
		SPRINT 3 TOTAL STORY POINTS - 12				
9	4	As a store manager, I want alerts for frequetly bought-together itrems that are placed far apart so that layout issues can be corrected.	6	Must do	High	6
4	4	As a buisness owner, I want reccomendations ranked by profits so that I don't waste time on low value associations.	5	Could do	Moderate	6
10	4	As a store manager, I want indicators about current trends so that we can anticipate demand changes and prepare beforehand.	5	Should do	High	1,7
		SPRINT 4 TOTAL STORY POINTS - 16				

ASSIGNMENT 5

	Name	Duration	Start	Finish	Predecessors	Resource Names
1	Problem Identification and Pr	1.667 days	3/13/26, 8:00 AM	3/16/26, 2:20 PM		Sneha[50%];Bandana[20%...
F1	Requirement Gathering	2 days	3/16/26, 2:20 PM	3/18/26, 2:20 PM	1	Anjali
3	User Story development	2 days	3/18/26, 2:20 PM	3/20/26, 2:20 PM	2	Bandana
4	SRS Documentation	4 days	3/20/26, 2:20 PM	3/26/26, 2:20 PM	3	Sneha
5	GUI Design(Wireframes and	4 days	3/26/26, 2:20 PM	4/1/26, 2:20 PM	4	Niyati
6	Level 0 and 1 DFD Diagrams	2 days	3/26/26, 2:20 PM	3/30/26, 2:20 PM	4	Mamta
7	Structure Chart Design	1 day	3/30/26, 2:20 PM	3/31/26, 2:20 PM	6	Mamta
8	Data Dictionary Preparation	2 days	3/30/26, 2:20 PM	4/1/26, 2:20 PM	6	Mamta[50%];Rishika[50%]
9	Database Design	2 days	3/30/26, 2:20 PM	4/1/26, 2:20 PM	6	Sneha[50%];Anjali[50%]
10	Algorithm Design	3 days	4/1/26, 2:20 PM	4/6/26, 2:20 PM	9	Bandana[50%];Ishaani[50%]
11	System Implementation	4 days	4/6/26, 2:20 PM	4/10/26, 2:20 PM	5;10	Niyati
12	Code Testing	3.333 days	4/10/26, 2:20 PM	4/15/26, 5:00 PM	11	Sneha[50%];Anjali[30%];Is...
13	System Testing	3.333 days	4/10/26, 2:20 PM	4/15/26, 5:00 PM	5;10;11	Sneha[50%];Anjali[20%];B...
14	Final Documentation and rep	1 day	4/16/26, 8:00 AM	4/16/26, 5:00 PM	13	Mamta







		Name	RBS	Type	E-mail Address	Material Label	Initials	Group	Max. Units	Standard Rate	Overtime Rate
1		Sneha		Work			S		100%	$1000.00/hour	$1100.00/hou
2		Anjali		Work			A		100%	$900.00/hour	$1000.00/hou
3		Bandana		Work			B		100%	$800.00/hour	$900.00/hou
4		Niyati		Work			N		100%	$700.00/hour	$800.00/hou
5		Rishika		Work			R		100%	$600.00/hour	$700.00/hou
6		Mamta		Work			M		100%	$500.00/hour	$600.00/hou
7		Ishaani		Work			I		100%	$200.00/hour	$300.00/hou

Software Requirements Specification (SRS)

Market Basket Analysis System

1. Introduction

1.1 Purpose

Retail stores generate large amounts of transaction data every day. However, identifying useful patterns from this data manually can be difficult and time-consuming. Store managers and business owners often rely on experience or observation to decide which products should be placed together or promoted as bundles.

The purpose of the **Market Basket Analysis System** is to assist retailers by automatically analyzing transaction data and identifying relationships between products that are frequently purchased together. For example, customers buying bread may also buy butter, or customers buying chips may also buy soft drinks.

By identifying these patterns, the system helps businesses make better decisions related to:

- Product placement inside the store
- Promotional bundle creation
- Marketing strategies
- Inventory planning

This document describes the functional and non-functional requirements of the system and will serve as a guide for the design and development of the Market Basket Analysis System.

1.2 Scope

The Market Basket Analysis System is a web-based analytics tool designed for retail environments such as supermarkets, convenience stores, or departmental stores.

The system analyzes historical sales transactions to identify relationships between products and detect buying patterns. These insights can help retailers improve store organization and marketing strategies.

For example, if the system detects that customers frequently buy **coffee and biscuits together**, the store manager can place these products close to each other or create a promotional offer.

The system provides:

- A dashboard displaying key sales insights
- Tools to analyze frequently purchased product combinations
- Trend analysis to detect changes in buying behavior
- Downloadable reports for sharing insights with stakeholders

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
MBA	Market Basket Analysis
KPI	Key Performance Indicator
Association Rule	Relationship between products frequently purchased together
Support	Frequency with which a product combination appears in transactions
Confidence	Probability that product B is purchased when product A is purchased
Lift	Measure of how strongly two products are associated
Dashboard	Interface displaying summarized business insights

1.4 Document Overview

This document is organized into the following sections:

- Introduction – explains the purpose and scope of the system
- Overall Description – describes the system environment and user roles
- Functional Requirements – lists the operations the system must perform
- Non-Functional Requirements – describes system performance, usability, and security
- Data Requirements – outlines the data required for analysis
- Constraints and Future Enhancements – discusses system limitations and possible improvements

2. Overall Description

2.1 Product Perspective

The Market Basket Analysis System is designed as a standalone web application that processes retail transaction data and presents analytical insights through dashboards and reports.

The system uses historical sales records to analyze how customers purchase products. By applying association analysis techniques, the system identifies relationships between products and highlights patterns that may not be obvious through manual observation.

The system architecture consists of three main layers:

1. User Interface Layer – allows users to interact with dashboards and reports

2. Application Logic Layer – processes transaction data and performs analysis
3. Database Layer – stores transaction and product information

This layered approach helps keep the system organized, scalable, and easy to maintain.

2.2 Product Functions

The system provides several analytical and reporting functions that help retailers understand customer purchasing behavior.

These functions include:

- Identifying products that are frequently purchased together
- Detecting emerging and seasonal buying patterns
- Displaying sales insights through visual dashboards
- Generating recommendations for product bundling
- Allowing users to filter transaction data for analysis
- Generating downloadable reports for business decision making

These features help businesses improve store layout, optimize promotions, and increase sales.

2.3 User Classes and Characteristics

Business Owner

The business owner mainly focuses on strategic decision-making and overall business performance.

Typical activities include:

- Monitoring overall sales performance
- Identifying profitable product combinations
- Observing long-term purchasing trends
- Planning promotional campaigns based on insights

The business owner primarily interacts with dashboards and recommendation insights.

Store Manager

The store manager focuses on day-to-day store operations.

Typical activities include:

- Identifying frequently purchased product combinations
- Improving store layout based on buying patterns

- Monitoring product placement efficiency
- Generating operational reports

The store manager uses the system mainly for operational analysis and reporting.

Marketing Executive

The marketing executive focuses on promotional strategies and campaigns.

Typical activities include:

- Identifying trending products
- Detecting seasonal product combinations
- Creating promotional bundles
- Suggesting add-on products for marketing offers

The marketing executive primarily uses trend insights and recommendation modules.

2.4 Operating Environment

The system will operate as a web application accessible through common web browsers.

Hardware Requirements

- Standard personal computer or laptop
- Internet connectivity

Software Requirements

- Web browser (Chrome, Firefox, Edge)
- Backend processing environment
- Database management system such as MySQL or PostgreSQL

Deployment Environment

The system can be deployed either on a local server or a cloud-based server environment.

2.5 Design Constraints

The following constraints may influence system design:

- The system requires structured transaction data.
- Very large datasets may affect processing speed.
- The accuracy of insights depends on the quality of the data provided.
- Appropriate security mechanisms must be implemented to protect business data.

2.6 Assumptions and Dependencies

The system assumes that:

- Retail transaction data will be available in structured format.
- Users have basic familiarity with reading charts and reports.

3. Functional Requirements

3.1 User Authentication

FR1 – The system shall allow users to log in using an email address and password.

FR2 – The system shall provide role-based access for different types of users.

FR3 – After successful login, the system shall redirect users to their respective dashboards.

3.2 Dashboard Module

FR4 – The system shall display a dashboard summarizing key sales insights.

FR5 – The dashboard shall include key metrics such as total sales and top-selling products.

FR6 – The system shall display charts representing buying patterns and trends.

FR7 – Users shall be able to navigate from the dashboard to detailed analysis pages.

3.3 Association Analysis Module

FR8 – The system shall analyze transaction data to identify frequently purchased product combinations.

FR9 – The system shall generate association rules including support, confidence, and lift values.

FR10 – The system shall display association results in tabular format.

3.4 Recommendation Module

FR11 – The system shall generate product bundle recommendations.

FR12 – Recommendations shall be ranked based on profitability or association strength.

FR13 – The system shall display explanations for generated recommendations.

3.5 Reporting Module

FR14 – The system shall allow filtering of reports by date range.

FR15 – The system shall allow exclusion of out-of-stock products from analysis.

FR16 – The system shall display a report preview before downloading.

FR17 – Users shall be able to download reports in CSV format.

FR18 – Users shall be able to download reports in PDF format.

3.6 Trend Analysis Module

FR19 – The system shall detect emerging buying patterns.

FR20 – Users shall be able to compare product associations across different time periods.

FR21 – The system shall identify seasonal product combinations.

FR22 – The system shall provide indicators that help anticipate demand changes.

4. Non-Functional Requirements

4.1 Performance Requirements

The system should process transaction datasets efficiently and provide analysis results within a reasonable time.

Dashboard loading and report generation should occur within a few seconds under normal conditions.

4.2 Security Requirements

User authentication must be implemented to prevent unauthorized access.

User credentials should be securely stored.

Role-based access control should ensure that users only access features relevant to their role.

4.3 Usability Requirements

The system interface should be simple and easy to understand.

Charts and tables should be clearly labeled to help users interpret insights easily.

Minimal training should be required to operate the system.

4.4 Reliability Requirements

The system should provide accurate analysis results based on the input data.

The system should maintain stable performance during continuous use.

Stored transaction data should remain secure even in the event of system failure.

4.5 Scalability Requirements

The system should be able to handle increasing amounts of transaction data without major system redesign.

Future versions of the system should support integration with larger retail systems.

5. Data Requirements

The system database will store the following information:

- Transaction ID
- Product ID
- Product Name
- Product Category
- Quantity Purchased
- Purchase Date
- Store Section
- Stock Availability
- Profit Margin

These attributes are necessary for performing market basket analysis and generating meaningful insights.

6. Constraints

System effectiveness depends on the availability and accuracy of transaction data.

Incomplete or incorrect data may affect the quality of generated insights.

The system performance may vary depending on the size of the dataset.

7. Future Enhancements

Future versions of the system may include:

- Real-time transaction analysis
- Integration with retail POS systems
- Mobile application support
- Advanced predictive analytics

8. Conclusion

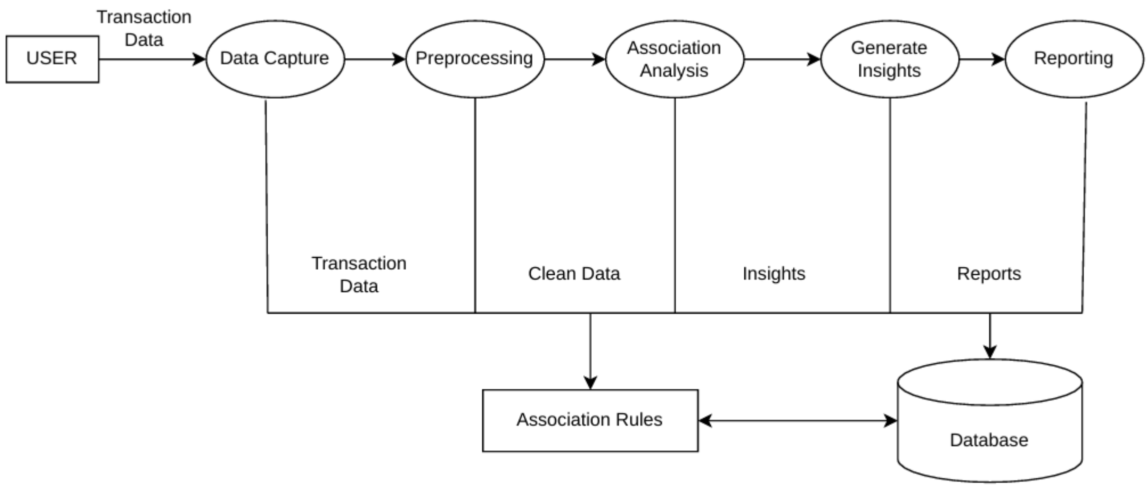
The Market Basket Analysis System is designed to help retailers better understand customer purchasing behavior through data analysis.

By identifying frequently purchased product combinations and emerging trends, the system enables businesses to improve store layout, plan better promotions, and optimize marketing strategies.

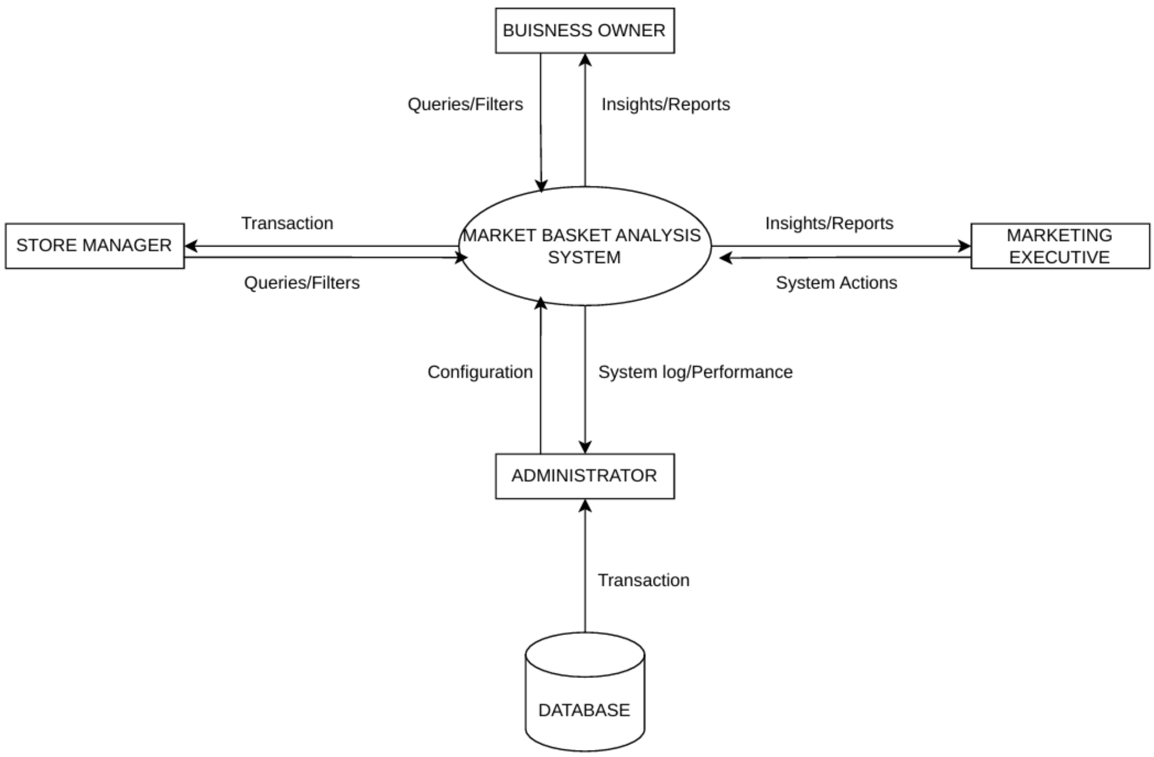
This document outlines the system requirements and serves as the foundation for the design and development of the Market Basket Analysis System.

DFDs, Structure Chart and Data Dictionary

Level 0 :



Level 1 :



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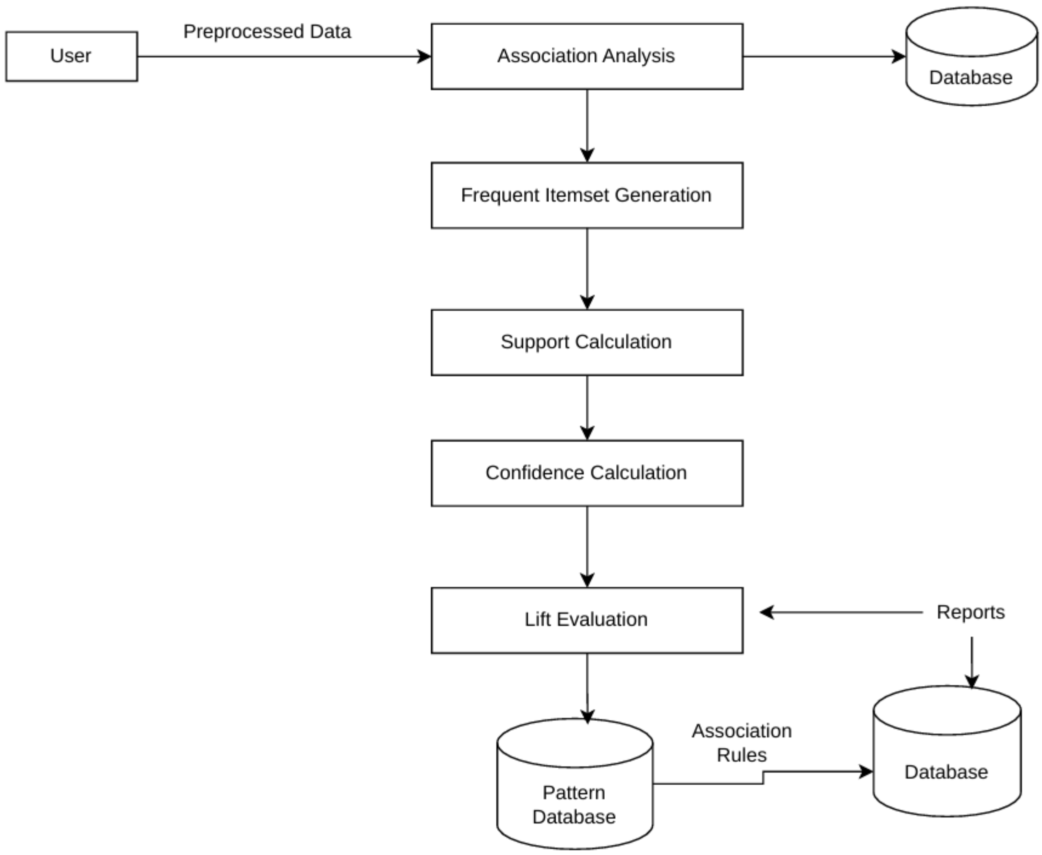
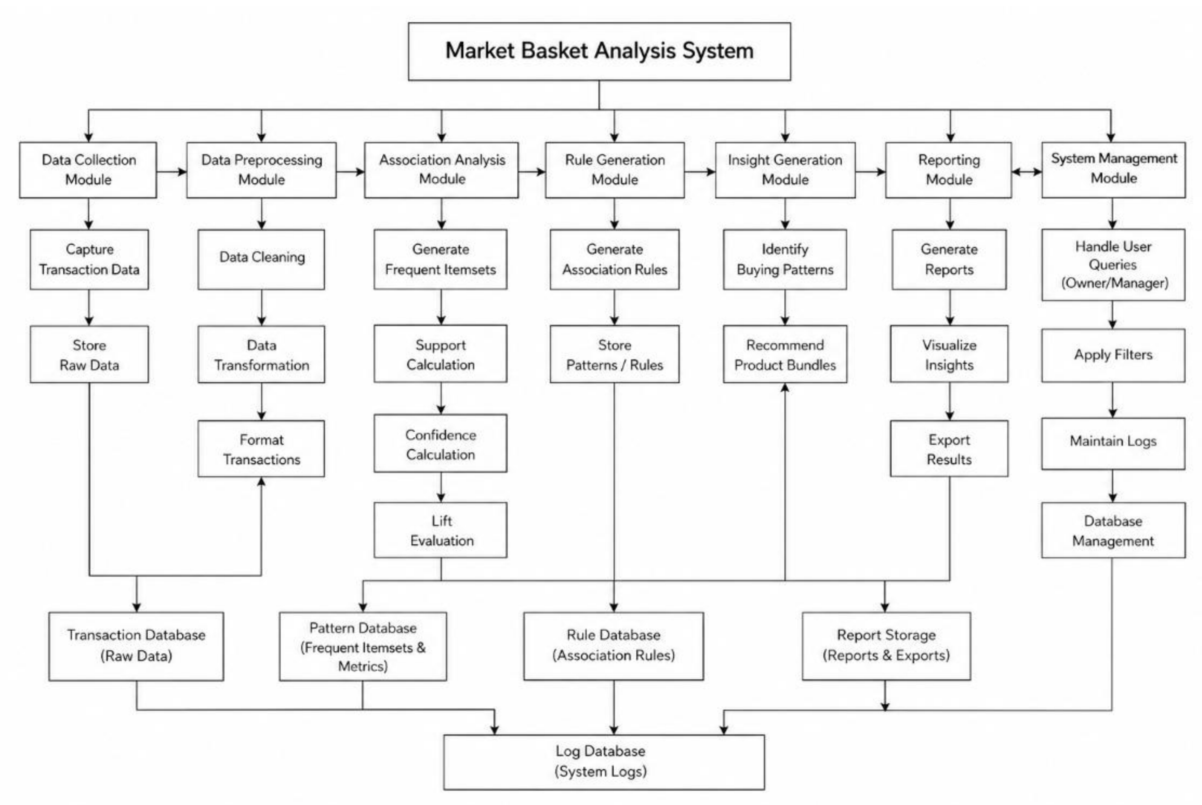


Structure Diagram :



Data Dictionary

1. Data Elements :

Name	Type	Description
Transaction_ID	Integer	Unique transaction identifier
Product_ID	Integer	Unique product identifier
Product_List	List	List of items in transaction
Raw_Data	List	Unprocessed transaction data
Clean_Data	List	Cleaned data after preprocessing
Transformed_Data	List	Structured data for analysis
Itemset	Set	Group of frequently occurring items
Support	Float	Frequency of itemset
Confidence	Float	Probability of rule
Lift	Float	Strength of association
Rule	String	Association rule (X → Y)
Pattern_Data	Set	Stored frequent itemsets
Report	String	Generated report
Query	String	User input query
Filter	String	Applied filter conditions
Log	String	System activity record

2. Data Stores

Data Store	Description
Transaction Database	Stores raw transaction data
Pattern Database	Stores frequent itemsets
Rule Database	Stores association rules
Report Storage	Stores generated reports
Log Database	Stores system logs

3. Data Flows

Data Flow	Description
Transaction Data	Input from user/store
Clean Data	Output of preprocessing
Transformed Data	Structured data
Itemsets	Output of analysis
Metrics	Support, confidence, lift
Rules	Generated association rules
Insights	Buying patterns
Reports	Final output
Logs	System activity
Query	User request
Filtered Data	Data after filtering

4. Processes

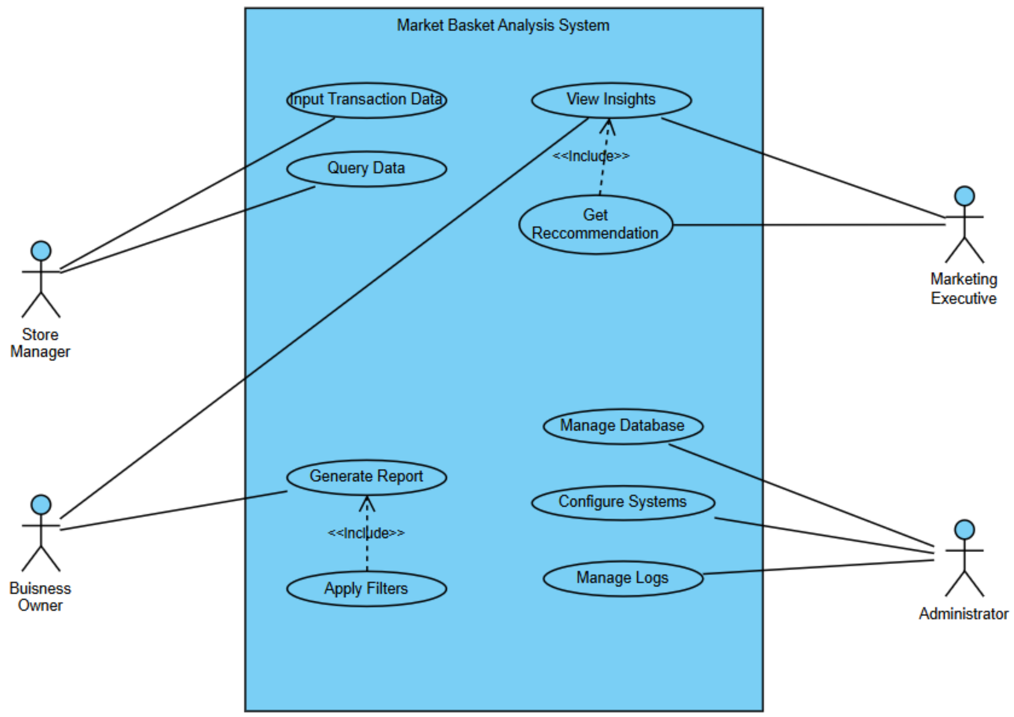
Process	Description
Capture Transaction Data	Collects transaction input
Store Raw Data	Saves raw data
Data Cleaning	Removes noise
Data Transformation	Converts format
Format Transactions	Prepares data
Generate Frequent Itemsets	Finds item patterns
Support Calculation	Calculates frequency
Confidence Calculation	Calculates probability
Lift Evaluation	Evaluates strength
Generate Association Rules	Creates rules
Store Patterns/Rules	Saves data
Identify Buying Patterns	Extracts insights
Recommend Product Bundles	Suggests combinations
Generate Reports	Creates reports
Visualize Insights	Displays insights
Export Results	Outputs results
Handle User Queries	Takes user input
Apply Filters	Filters data
Maintain Logs	Records activity
Database Management	Maintains DB

5. Input Structure

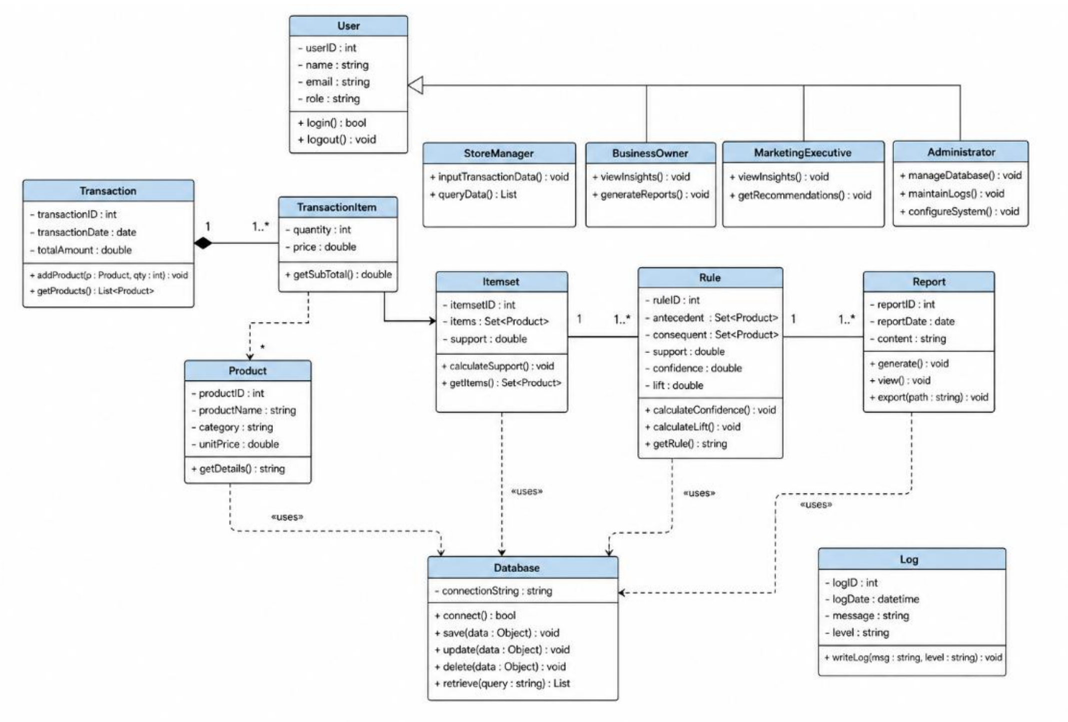
6. Output Structure

Output	Fields
Itemset Output	Itemset, Support
Rule Output	Rule, Confidence, Lift
Report Output	Report
Insight Output	Patterns, Recommendations
Log Output	Log

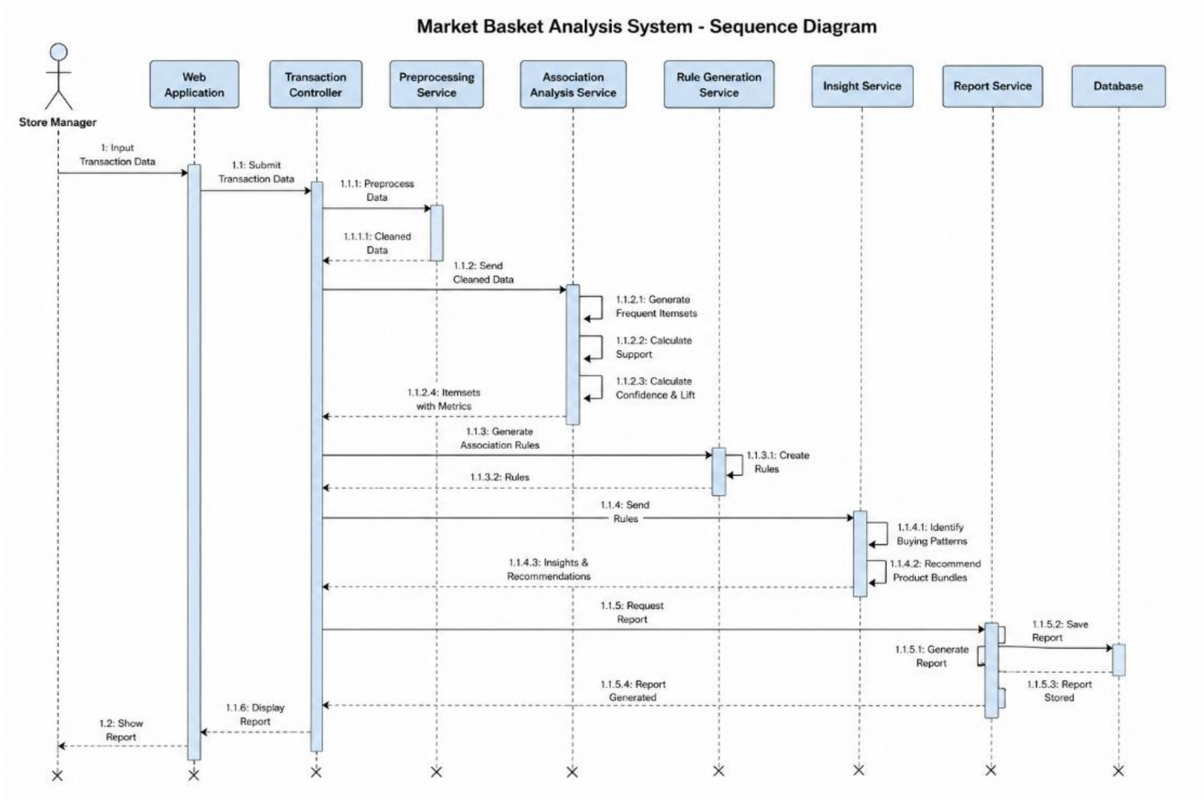
Use Case Diagram :



Class Diagram :



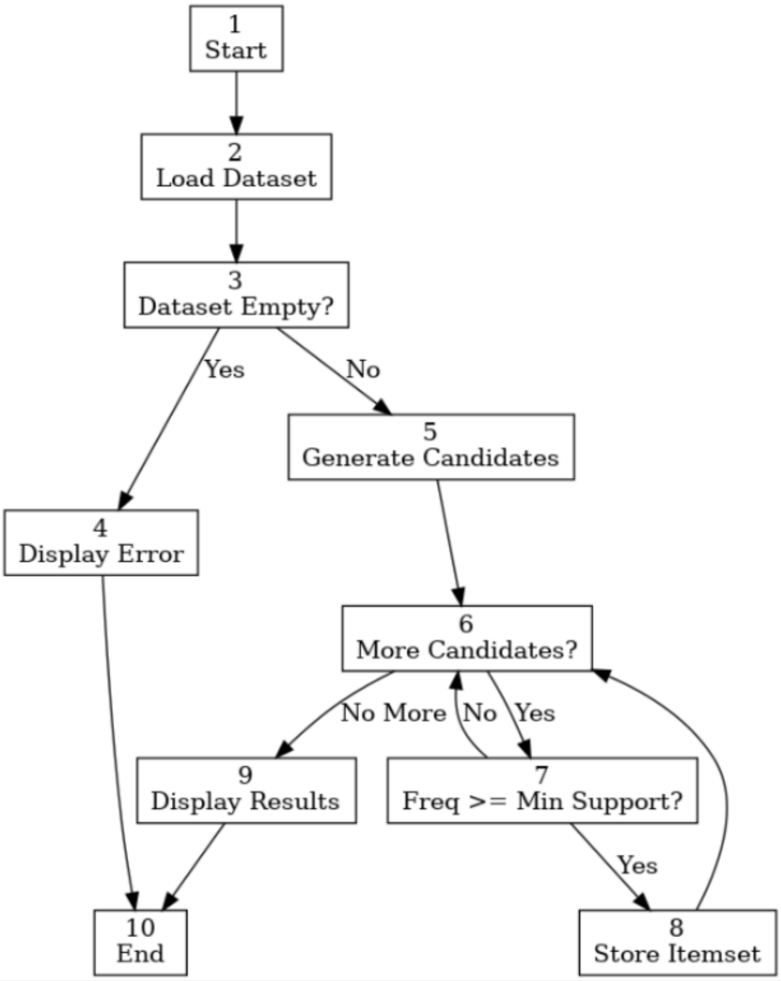
Sequence Diagram :



PSEUDO CODE, CFGs, LI PATHS AND TEST CASES

Frequent Itemset Generation

```
BEGIN

INPUT: transaction_data, min_support

CREATE empty dictionary item_count

FOR each transaction IN transaction_data DO

  FOR each item IN transaction DO

    IF item exists in item_count THEN

      item_count[item] = item_count[item] + 1

    ELSE

      item_count[item] = 1

    ENDIF

  ENDFOR

ENDFOR

CREATE empty list frequent_itemsets

FOR each item IN item_count DO

  support = item_count[item] / total_transactions

    IF support >= min_support THEN

    ADD item TO frequent_itemsets

  ENDIF

ENDFOR

OUTPUT frequent_itemsets

END
```



Cyclomatic Complexity

Formula 1 :

$$V(G) = E - N + 2$$

Nodes = 10

Edges = 11

$$V(G) = 11 - 10 + 2$$
$$V(G) = 3$$

Formula 2 :

$$V(G) = P + 1$$
$$V(G) = 2 + 1$$
$$V(G) = 3$$

Formula 3 :

Regions = 3

$$V(G) = 3$$

LI Paths

Path 1

1→2→3→4→10

Path 2

1→2→3→5→6→7→8→6→9→10

Path 3

1→2→3→5→6→7→6→9→10

<u>Test Cases</u>

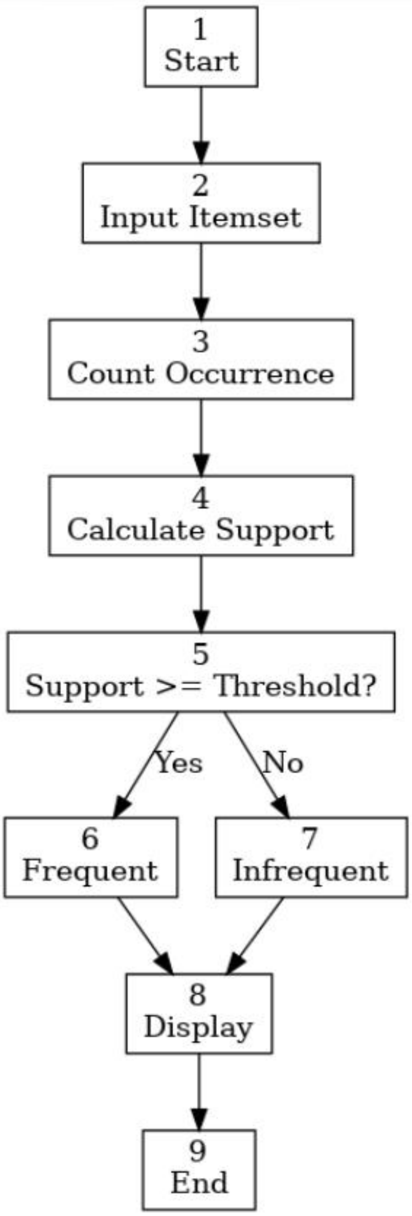
Feature	Test Case
Empty Dataset	Upload empty transaction file
Valid Frequent Itemset	Bread-Milk appears 50 times, Min Support=20
Non Frequent Itemset	Bread-Jam appears 3 times, Min Support=20

<u>Detailed Test Case Table</u>

Feature	Preconditions	Test Steps	Expected Result	Pass Criteria
Empty Dataset	Dataset uploaded	Run analysis	Error message displayed	No crash
Frequent Itemset Found	Support above threshold	Execute analysis	Itemset stored	Appears in output
Support Not Met	Support below threshold	Execute analysis	Itemset discarded	Does not appear

Support Calculation

```
BEGIN
INPUT: itemset, transaction_data
count = 0
total_transactions = number of transactions
FOR each transaction IN transaction_data DO
   IF itemset is subset of transaction THEN
      count = count + 1
   ENDIF
ENDFOR
support = count / total_transactions
OUTPUT support
END
```



Cyclomatic Complexity

Formula 1 :

Nodes = 9

Edges = 9

$$V(G) = 9 - 9 + 2$$
$$V(G) = 2$$

Formula 2

Predicate Nodes = 1

$$V(G) = 1 + 1 = 2$$

Formula 3

Regions = 2

$$V(G) = 2$$

LI Paths

Path 1

1→2→3→4→5→6→8→9

Path 2

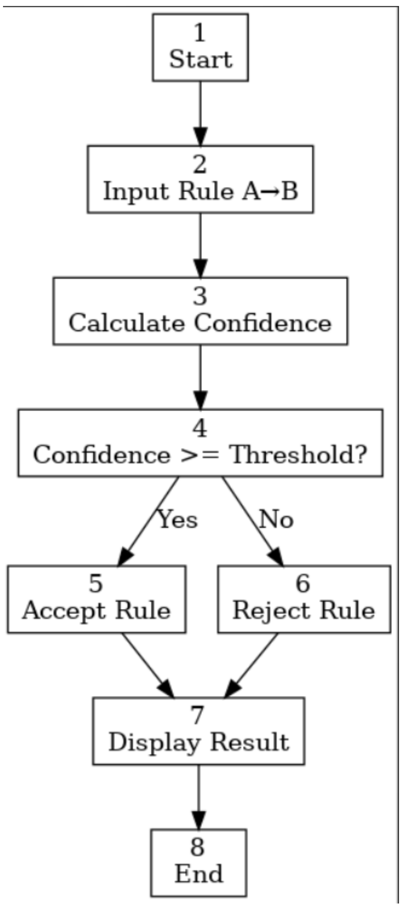
1→2→3→4→5→7→8→9

Test Cases

Feature	Preconditions	Test Steps	Expected Result	Pass Criteria
Frequent Itemset	Support=0.35, Threshold=0.2	Calculate support	Frequent	Marked Frequent
Infrequent Itemset	Support=0.05, Threshold=0.2	Calculate support	Infrequent	Marked Infrequent

Confidence Calculation

```
BEGIN
INPUT: itemset_X, itemset_Y, transaction_data
count_X = 0
count_XY = 0
FOR each transaction IN transaction_data DO
   IF itemset_X is subset of transaction THEN
      count_X = count_X + 1
            IF itemset_Y is also subset of transaction THEN
         count_XY = count_XY + 1
      ENDIF
   ENDIF
ENDFOR
confidence = count_XY / count_X
OUTPUT confidence
END
```



Cyclomatic Complexity

Formula 1

Nodes = 8

Edges = 8

$$V(G) = 8 - 8 + 2$$
$$V(G) = 2$$

Formula 2

Predicate Nodes = 1

$$V(G) = 1 + 1$$
$$V(G) = 2$$

Formula 3

Regions = 2

$$V(G) = 2$$

LI Paths

Path 1

1→2→3→4→5→7→8

Path 2

1→2→3→4→6→7→8

Test Cases

Feature	Preconditions	Test Steps	Expected Result	Pass Criteria
Valid Rule	Confidence=0.85, Threshold=0.7	Execute calculation	Rule accepted	Appears in result
Weak Rule	Confidence=0.30, Threshold=0.7	Execute calculation	Rule rejected	Not recommended